

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Problem solving skills, critical thinking
2. Should know the use of proper prompts
3. Should know system design and architecture

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A	Yes	Yes	Because a calculator mainly involves code to do calculation
B	Yes	Yes	Because payroll requires analysis, time calculation and maintenance which involves software
C	Yes	No	Because AI cant do testing and analysis by itself.

Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Management	Reports, workload management
Practitioners	Patient history, schedule
Patients	Reliable appointment, reminders, edit appointment
IT support	Maintaining the software, edit information when needed.

Task 2 - Identify current problems

1. Current paper-based causes errors and limited edits
2. Double booking and improper handling of schedule can occur
3. Slow access to patient data
4. Less secured and chances of loss of data

Task 3 - Ask client questions

1. Do you need auto reminders or not?
2. Should patient data be shared?
3. What type of question should be asked during appointments?
4. How many staff will use the system at the same time?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes, the client wants to improve appointment handling.	Yes — core requirements.	Include
Facial recognition login	No- client never mentioned biometrics.	No — too advanced, privacy concerns.	Exclude
AI diagnosis recommendations	No — Clients only want admin improvements.	No — medical risk, legal issues.	Exclude.
Patient search	Yes — paper records make finding info slow.	Yes — supports core workflow.	Include.
Online payment	No — client did not mention billing or payments.	Probably no — unrelated to appointment management.	Exclude unless client requests.
Practitioner schedule view	Yes — needed for booking and avoiding	Yes — directly supports appointments.	Include.

	conflicts.		
Insurance processing	No — client did not mention insurance workflows.	No — large, complex, out of scope.	Exclude.
Treatment-plan generation	No — client wants admin improvements, not clinical decision support.	No — high-risk medical features.	Exclude.

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

- A software engineer must validate the requirements of the client because only humans have the ability to understand the real world needs. They have the ability to judge the social and legal obligations, constraints, user needs as well as organizational context. AI has the ability to generate code, but it cannot verify user needs. Also, AI cannot reliably judge whether the solution is appropriate or not. Also, there are constraints like safety and ethics aligned with the client's actual goal.